

Week 17

Integrating Open Data and Machine Learning Project Skills

North Metropolitan TAFE – Certificate IV Information Technology
Data Science and AI — Week 17

Learning Objectives

- Synthesize dataset sourcing, analysis, and machine learning skills into a comprehensive project.
- Apply best practices for project integration, review, and peer feedback.
- Finalize and critically assess project work using industry-standard tools and workflows.
- Prepare for final assessment submission by aligning with competency and industry requirements.

Session Agenda

- Recap of previous advanced concepts and connection to current week.
- Full project integration: bringing open data and ML skills together.
- Practical review cycles and structured peer feedback.
- Finalizing project documentation and assessment readiness.
- Q and A plus next steps for final course synthesis.

Review: Building to Synthesis

- Weeks 14 to 16 covered; intermediate/advanced machine learning, scaling data analysis, and advanced ethics.
- Skills covered; unsupervised ML techniques, handling large-scale open datasets, ethical and privacy challenges.
- This week; connect all prior learning into your capstone project for assessment submission.

Why Integration Matters

- Real-world AI and data science require end-to-end workflows; open data sourcing, cleaning, ML modelling, and transparent reporting.
- Employers value ability to manage projects holistically; not just run isolated tools.
- Integrated project demonstrates mastery of course competencies and industry relevance.

Project Integration Steps

- Revisit project scope and objectives; ensure clear problem statement.
- Optimize data sourcing, cleaning, and preprocessing; apply data ethics and documentation standards.
- Version and document all changes; ensure reproducibility.
- Apply and tune ML models; interpret results for real-world cases.
- Compile all work into a clear, industry-ready report or notebook.
- Critically review your pipeline for gaps, bias, or reproducibility issues.

Peer Review and Feedback

- Peer review surfaces blind spots, errors, or new insights; essential in industry and research.
- Guidelines; be specific, constructive, and respectful. Identify both strengths and suggested improvements.
- Use rubric aligned to assessment criteria for structured peer feedback.

Example Workflow; Image Classification Pipeline

- Example; Open-source dataset (e.g., COCO), data cleaning in pandas and numpy, versioning with DVC.
- ML pipeline; feature extraction with CLIP, model training with scikit-learn or PyTorch.
- Visualize results in matplotlib; report reproducibility and ethics considerations in Jupyter Notebook.
- Final output; industry-standard project ready for peer and lecturer review.

Hands-on Activity; Review Cycle and Reflection

- Swap projects with a peer; conduct review using provided rubric.
- Complete review form; highlight key strengths, ethics compliance, and areas for improvement.
- Revise your work based on feedback; reflect and document learnings in project log.
- Arrange short feedback discussion to clarify points and share insights.

Assessment and Submission Preparation

- Projects are assessed for; critical research, EDA, practical ML application, and knowledge evidence per unit standards.
- Ensure all parts are documented, reproducible, and clearly evidence ethical and responsible data practices.
- Use industry templates and documentation standards; align with example projects discussed in class.
- Submit all required files; notebook or report, data pipeline scripts, and peer review reflection.

Common Pitfalls and Best Practices

- Incomplete documentation or lack of reproducibility is the most frequent reason for reassessment.
- Avoid superficial analysis; provide in-depth discussion of results and limitations.
- Best practice; cross-check project against grading rubric and minimum requirements before final submission.

Summary and Next Steps

- Week 17 brings together all major skills for comprehensive demonstration and final review.
- Seek clarification or support if any aspect of your project is unclear or incomplete.
- Next week; final assessment prep, knowledge review, and course synthesis.
- Reminder; reassessment period is Weeks 19 to 20 if needed.

